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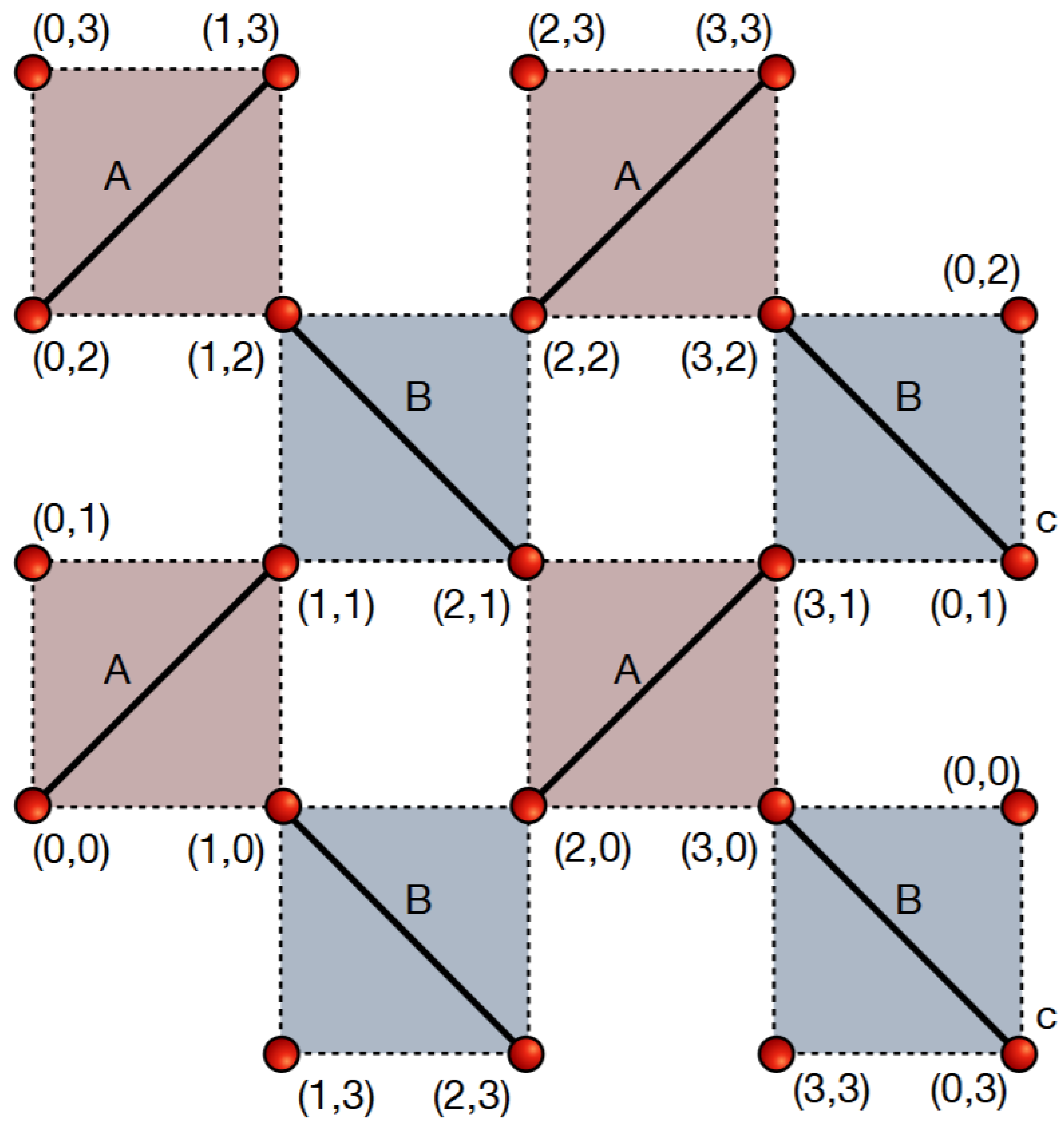

Projected Density Matrix Sampling on the Shastry-Sutherland Model

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Max Planck Graduate Center for Quantum Materials

Selection Symposium

Quantum Monte Carlo and The Sign Problem

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- Computing the **low-lying spectrum** remains a frontier in Quantum Many-Body Physics.
- The fundamental difficulty is the **exponential growth of the Hilbert space** dimension with the quantum system size, which makes Exact Diagonalization (ED) intractable.
- **Quantum Monte Carlo (QMC)** methods are one of the most powerful approaches to tackle this.

Two Difficulties!

The Sign Problem

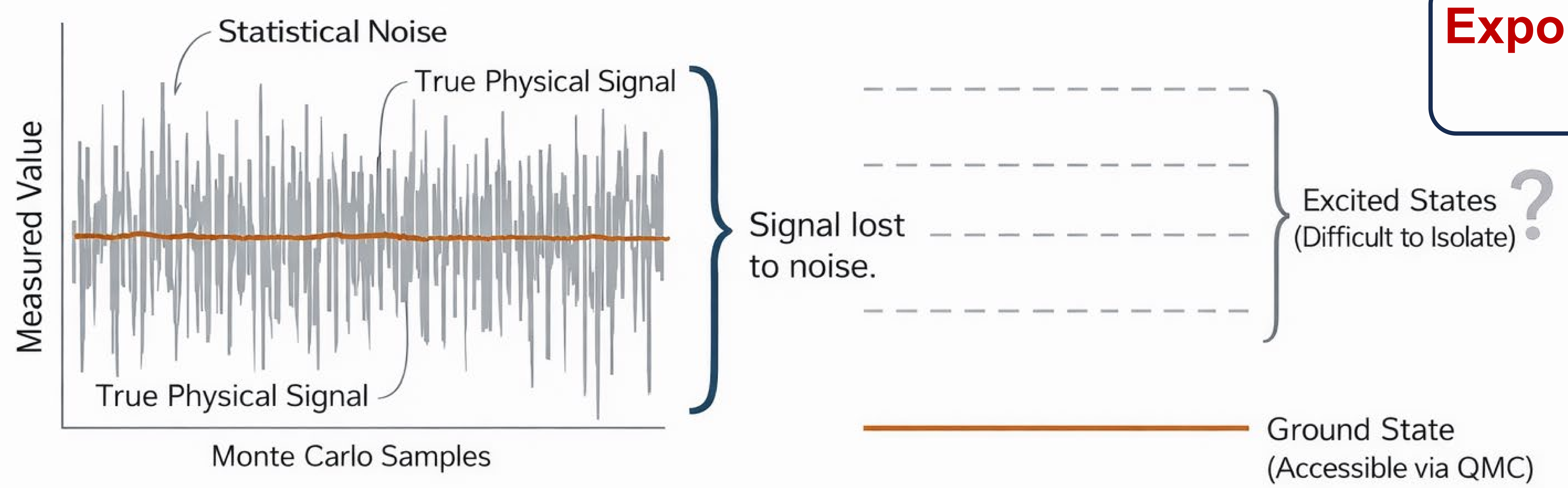
The notorious sign problem makes Monte Carlo sampling impossible for many important systems, as weights are not always positive. In its general form, this problem is computationally hard.

Accessing Excited States

Standard QMC methods are designed to find the ground state. Extracting the spectrum of low-lying excited states is exceptionally difficult.

NP-hard in general!

Exponential suppression ~ ΔE as $Z = Tr(e^{-\beta H})$!



The Projected Density Matrix Sampling (PDMS) Method

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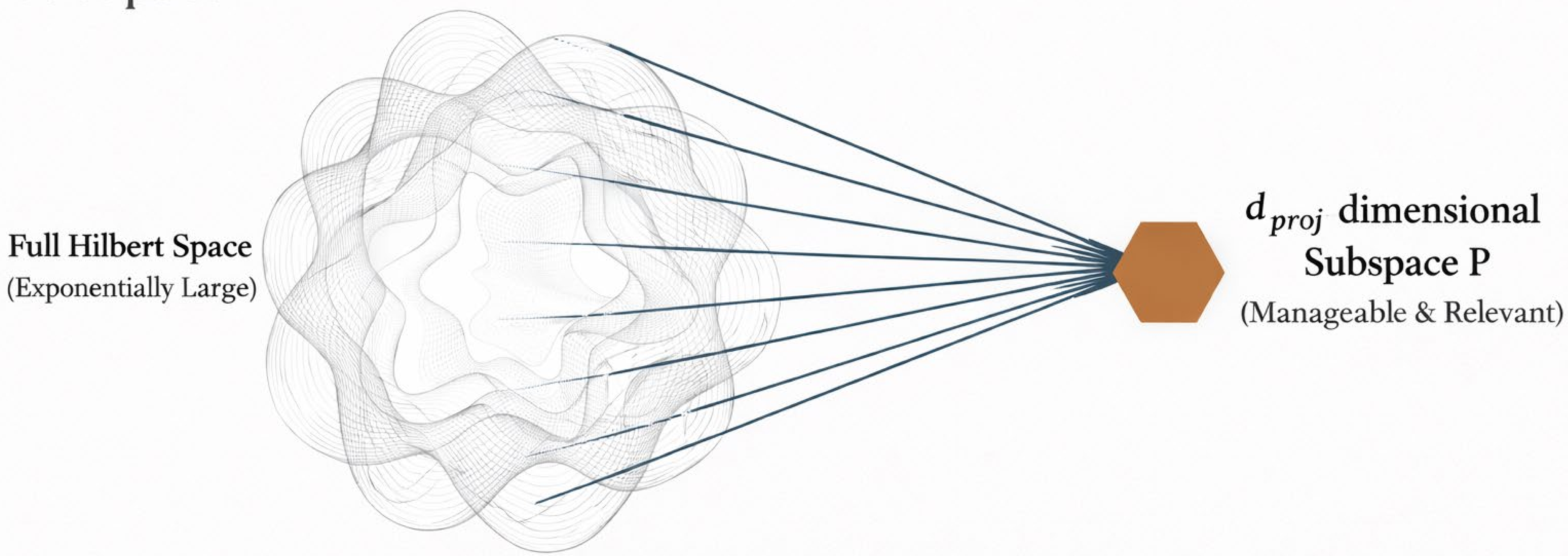
Instead of tackling the entire, exponentially large Hilbert space, we project the problem onto a much smaller, physically-motivated subspace.



PDMS isolates the **low-energy spectrum** by projecting the thermal matrix, $\rho = e^{-\beta H}$, onto a **judiciously chosen** subspace of states.

Motivating Question: What if we converge to the low-energy sector before the sign problem becomes prohibitive?

The Projected Density Matrix Sampling (PDMS) Method

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The Framework: Extracting Energies from Projected Matrices

Key idea: We define two $d_{proj} \times d_{proj}$ matrices by projecting the thermal density matrix and the thermal energy onto the chosen subspace P, spanned by states $|\psi_a\rangle$.

Projected Density Matrix (Z)

$$Z_{ac} = \langle \psi_a | e^{-\beta H} | \psi_c \rangle$$

Represents the overlaps between subspace states after evolution in imaginary time.

Projected Energy Matrix (E)

$$E_{ac} = \langle \psi_a | H e^{-\beta H} | \psi_c \rangle$$

Represents the energy expectation value within the projected, evolved subspace.

The Result

$$EZ^{-1}$$

The d_{proj} eigenvalues of this matrix product are the projected thermal energies, $e_a(\beta)$.

In the zero-temperature limit (large β), these energies converge to the d_{proj} lowest eigenvalues of the Hamiltonian H that have overap with the subspace P: $e_a(\beta \to \infty) \to E_a$.

Motivating Question: What if we converge to the low-energy sector before the sign problem becomes prohibitive?

The Continuous-Time Worldline Representation

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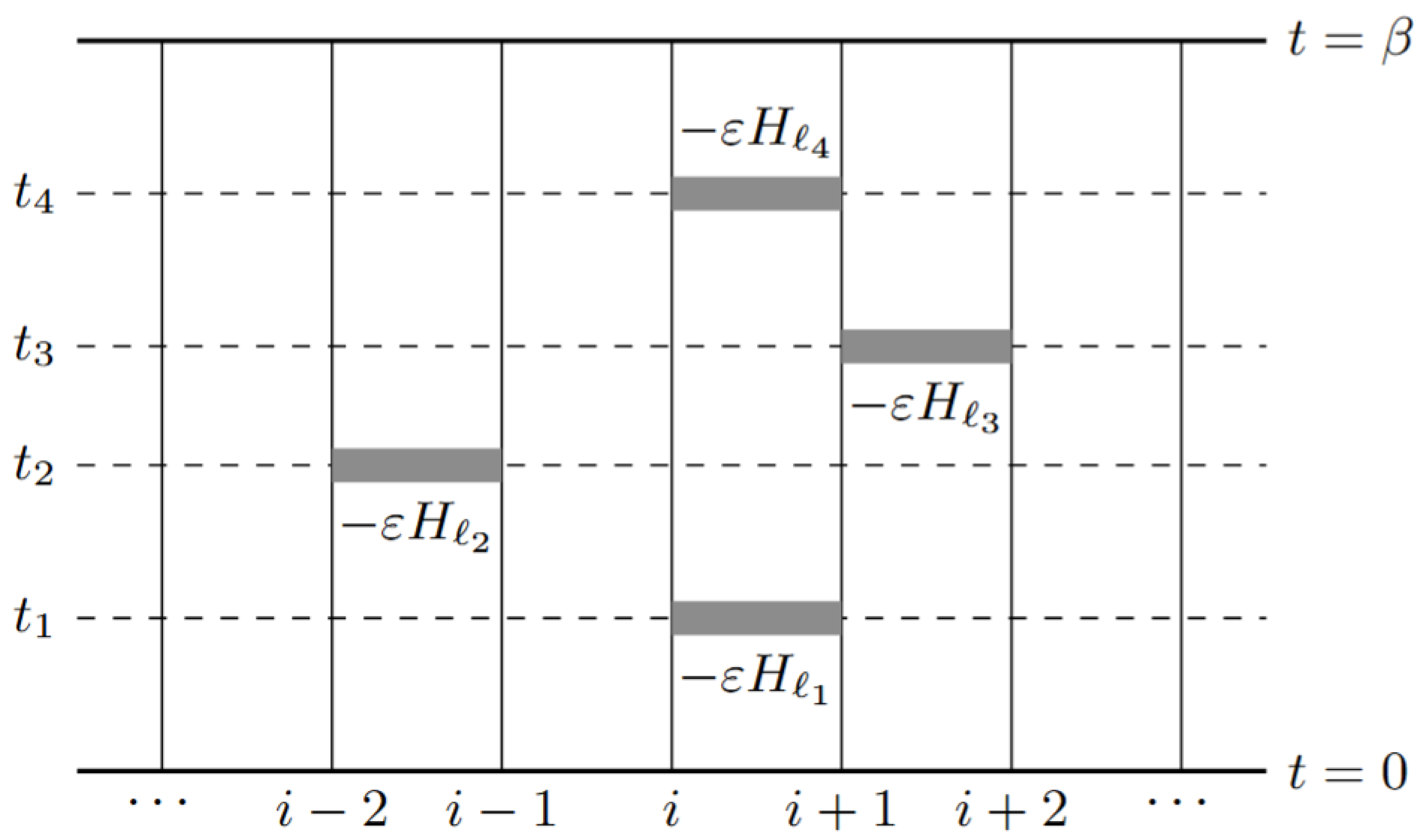
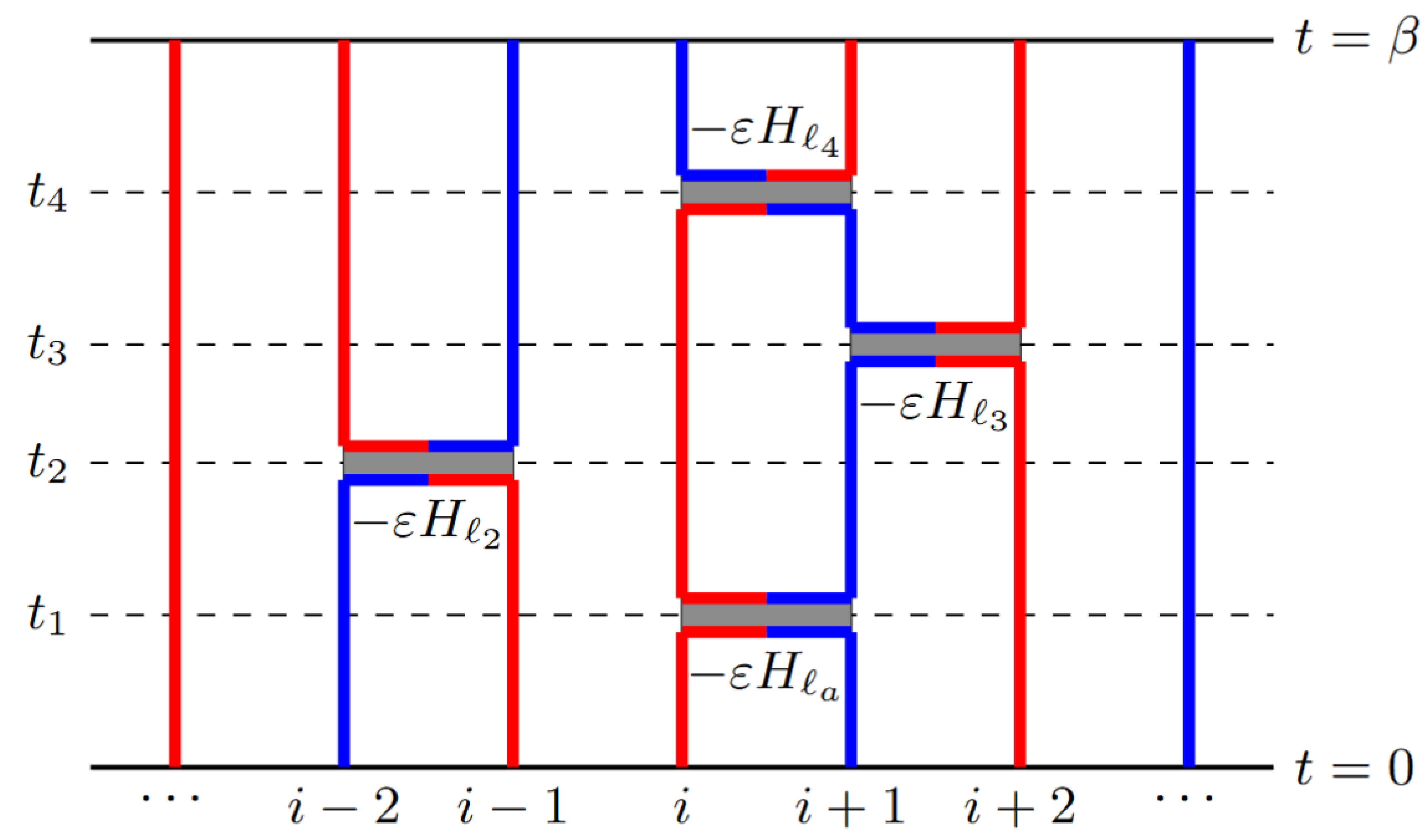
$$Z_{ij} = \sum_{i,j} \langle \psi_i | e^{-\beta H} | \psi_j \rangle = \lim_{\substack{L_t \to \infty \\ \epsilon \to 0,\, L_t \epsilon = \beta}} \sum_{i,j} \langle \psi_i | \left(e^{-\epsilon H} \right)^{L_t} | \psi_j \rangle$$

$$= \sum_{i,j} \langle \psi_i | \left(1 - \epsilon H \right)^{L_t} | \psi_j \rangle = \sum_{\mathcal{C}} W(\mathcal{C})\, S(\mathcal{C})$$

$$E_{ij} = \sum_{i,j} \langle \psi_i | H\, e^{-\beta H} | \psi_j \rangle = \lim_{\substack{L_t \to \infty \\ \epsilon \to 0,\, L_t \epsilon = \beta}} \sum_{i,j} \langle \psi_i | H \left(e^{-\epsilon H} \right)^{L_t} | \psi_j \rangle$$

$$= \sum_{i,j} \langle \psi_i | H \left(1 - \epsilon H \right)^{L_t} | \psi_j \rangle = \sum_{\mathcal{C}} W(\mathcal{C})\, S(\mathcal{C})\, E(\mathcal{C})\,.$$





Ingredients for the PDMS Method

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The Workflow: Monte Carlo Evaluation in a Projected Space

Starting Point
Lattice Hamiltonian H

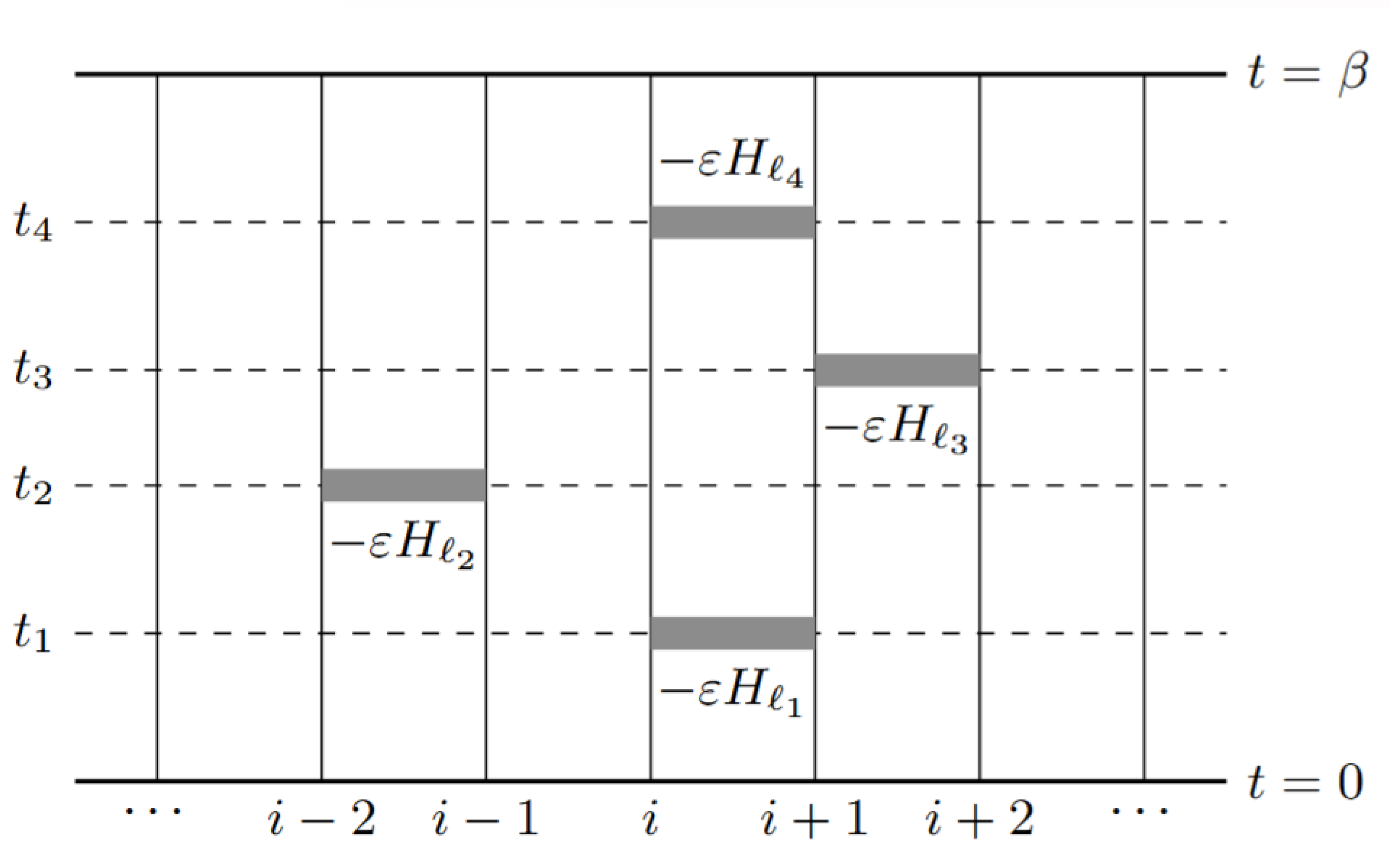
Worldline Representation
Trotterized, continuous-time worldline configuration

Monte Carlo Sampling
Stochastic updates: Bond & Spin Moves

Projection and Statistics
Projection Subspace P

Measurements
Extract Observables & Physical Quantities

Can equally well do in other representations such as SSE.



What is Geometric Frustration?

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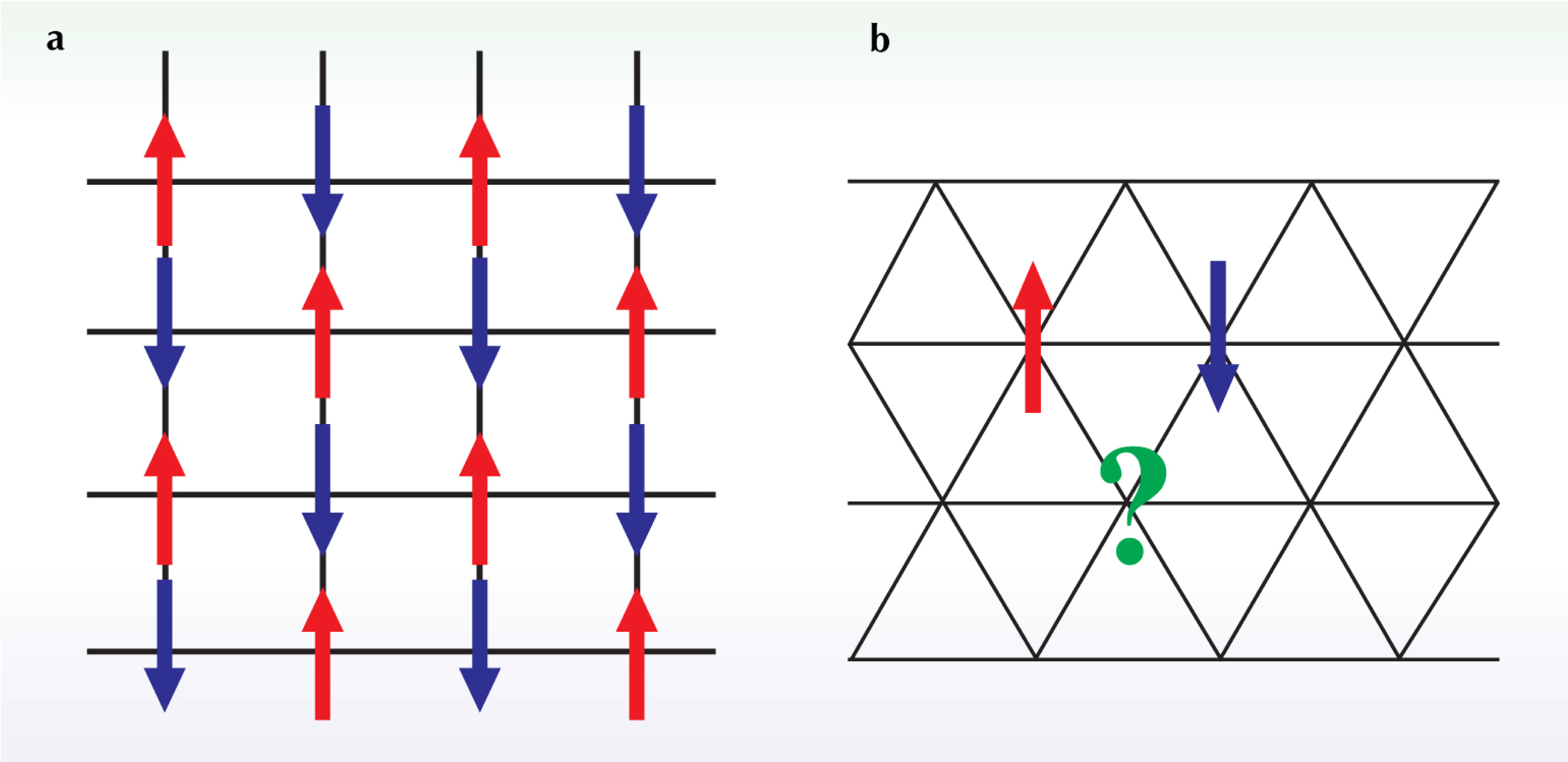

Figure 1. *<u>A geometrically frustrated system</u>* is one in which the geometry of the lattice precludes the simultaneous minimization of all interactions. (**a**) In the un-frustrated antiferromagnet on the square lattice, each spin can be anti-aligned with all its neighbors. (**b**) On a triangular lattice, such a configuration is impossible: Three neighboring spins cannot be pairwise anti-aligned, and the system is frustrated. (*Adapted from Physics Today. 2006;59(2):24-29. doi:10.1063/1.2186278*)

It is well-known that geometric frustration leads to the sign problem!

Geometric Frustration and the Origin of the Sign Problems

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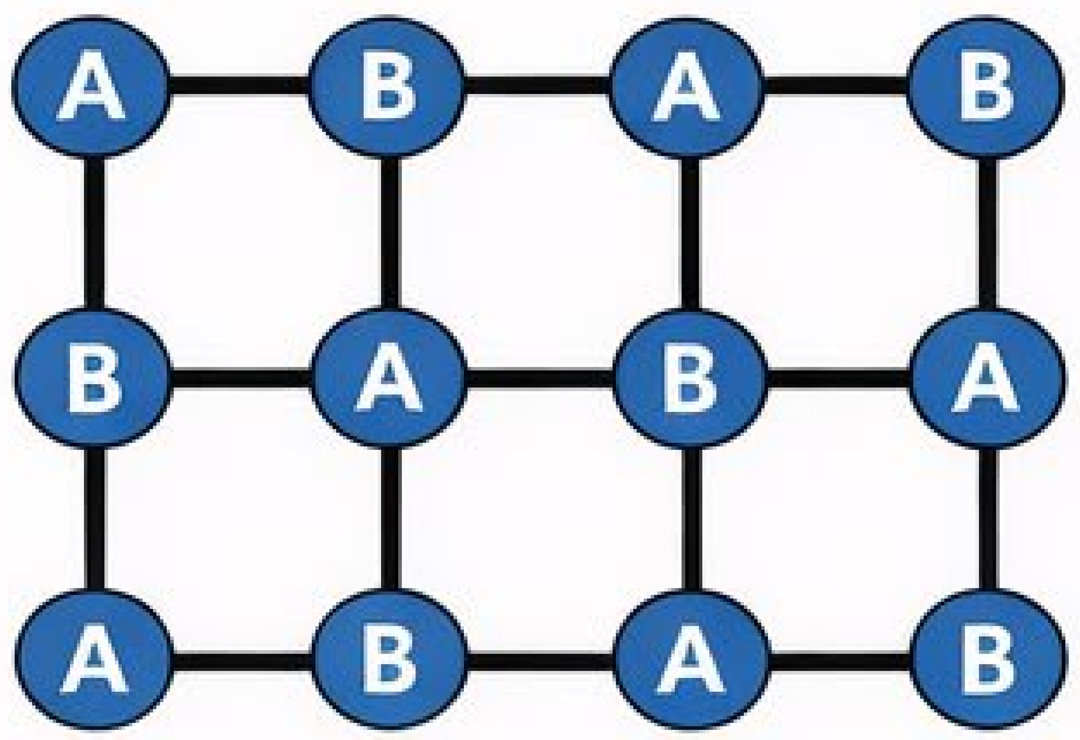
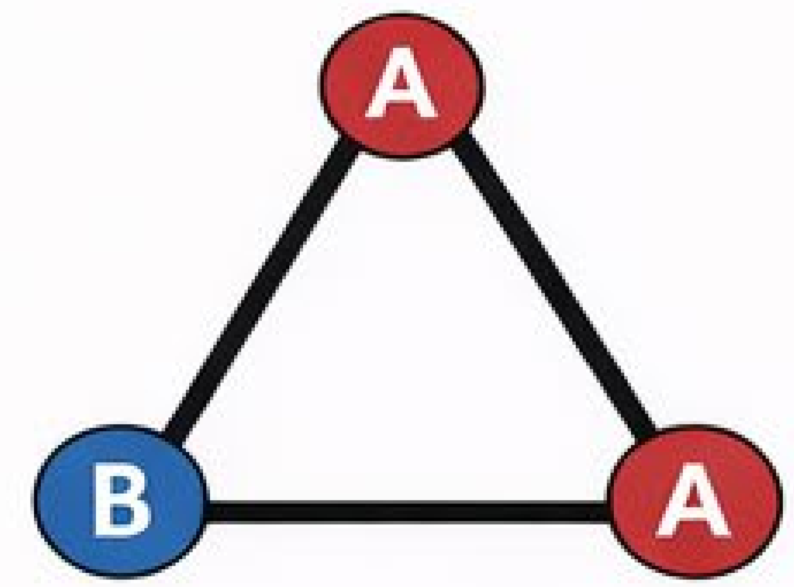
Unfrustrated lattice (bipartite)



Square lattice

Only even loops

Frustrated lattice (nonbipartite)



Triangular plaquette

Odd loop present

Marshall sign rule

Bipartite lattice:

→ Sublattice **rotation**

⇒ All matrix elements *nonpositive*

Breakdown under frustration

Frustrated lattice:

→ No consistent A/B labeling

⇒ **Negative** matrix elements

Key message:

Geometric frustration ⇒ Nonbipartite lattice ⇒ **Sign** problem in QMC.

Introduction: The Shastry-Sutherland Model

$$H^{\mathrm{ssm}} = J' \sum_{n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + J \sum_{n.n.n.} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

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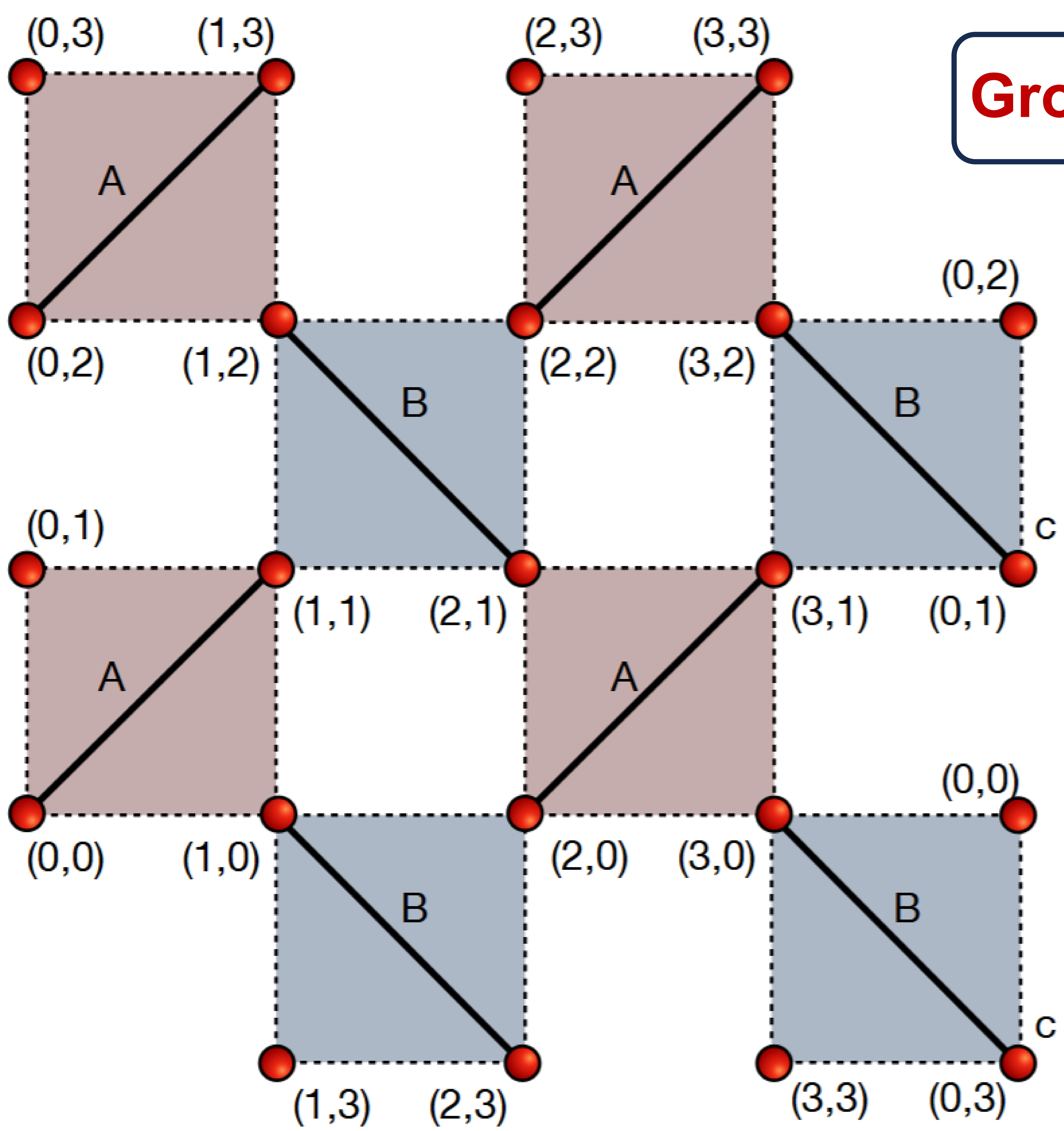
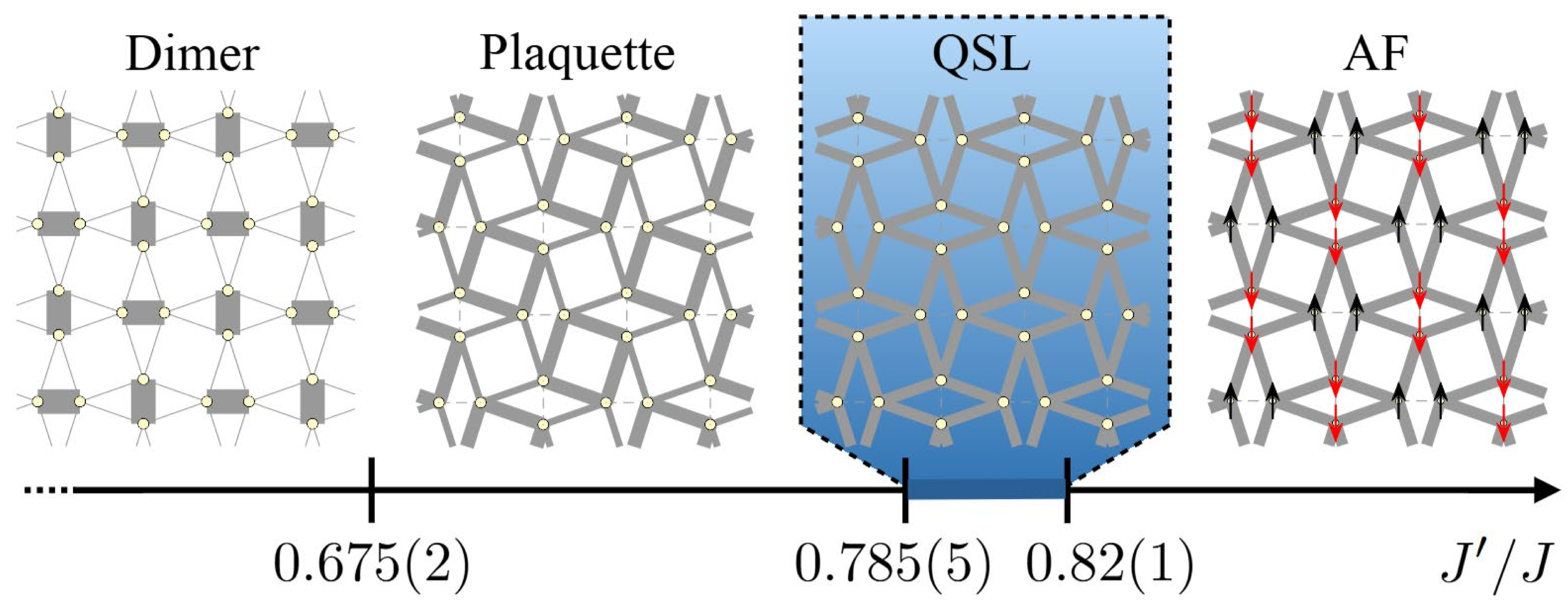


Ground state is Product of Dimer Singlets up to J'/J <=0.5.



(Adapted from arXiv:2502.14091).

Shastry-Sutherland paper in 1981…45 years ago!

Geometric Frustration to Sign Problem!

Figure 1. The 4x4 Shastry-Sutherland lattice (left) and its phase diagram (right).

Projection Subspaces Captures the Physics : I

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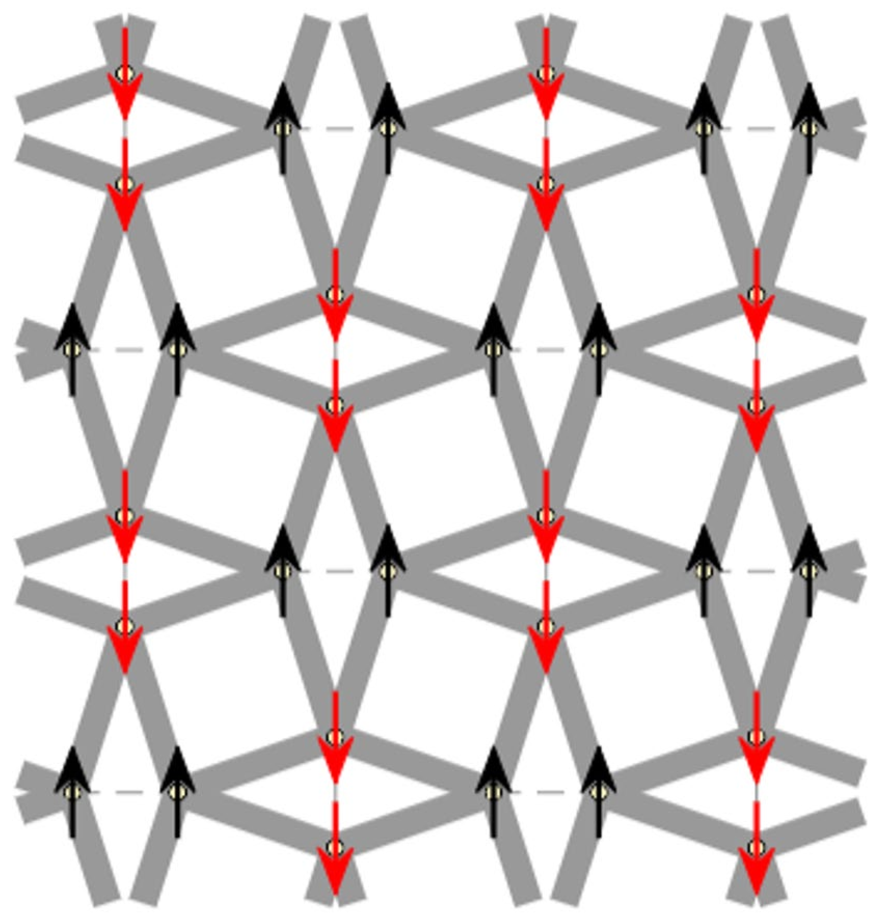


Our goal with PDMS is to capture the right physics via the projection!

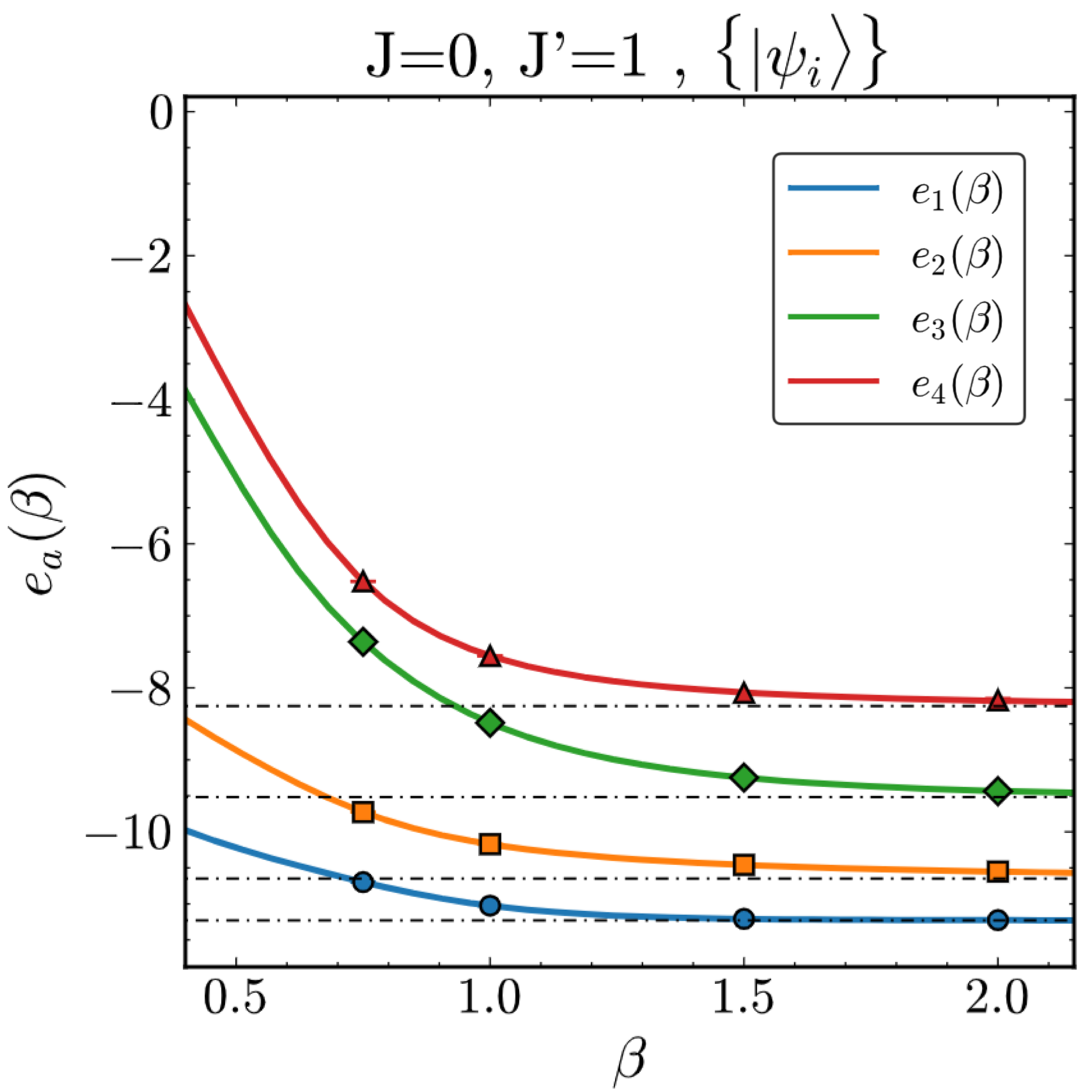
$$\hat{Q} = \prod_i (2s_i^x)$$

Two projection sectors with +1 and -1 eigenvalues!

P+ (psi states)

These states are motivated by the physics of the anti-ferromagnet!

4x4 SSM in the Unfrustrated Limit: Results



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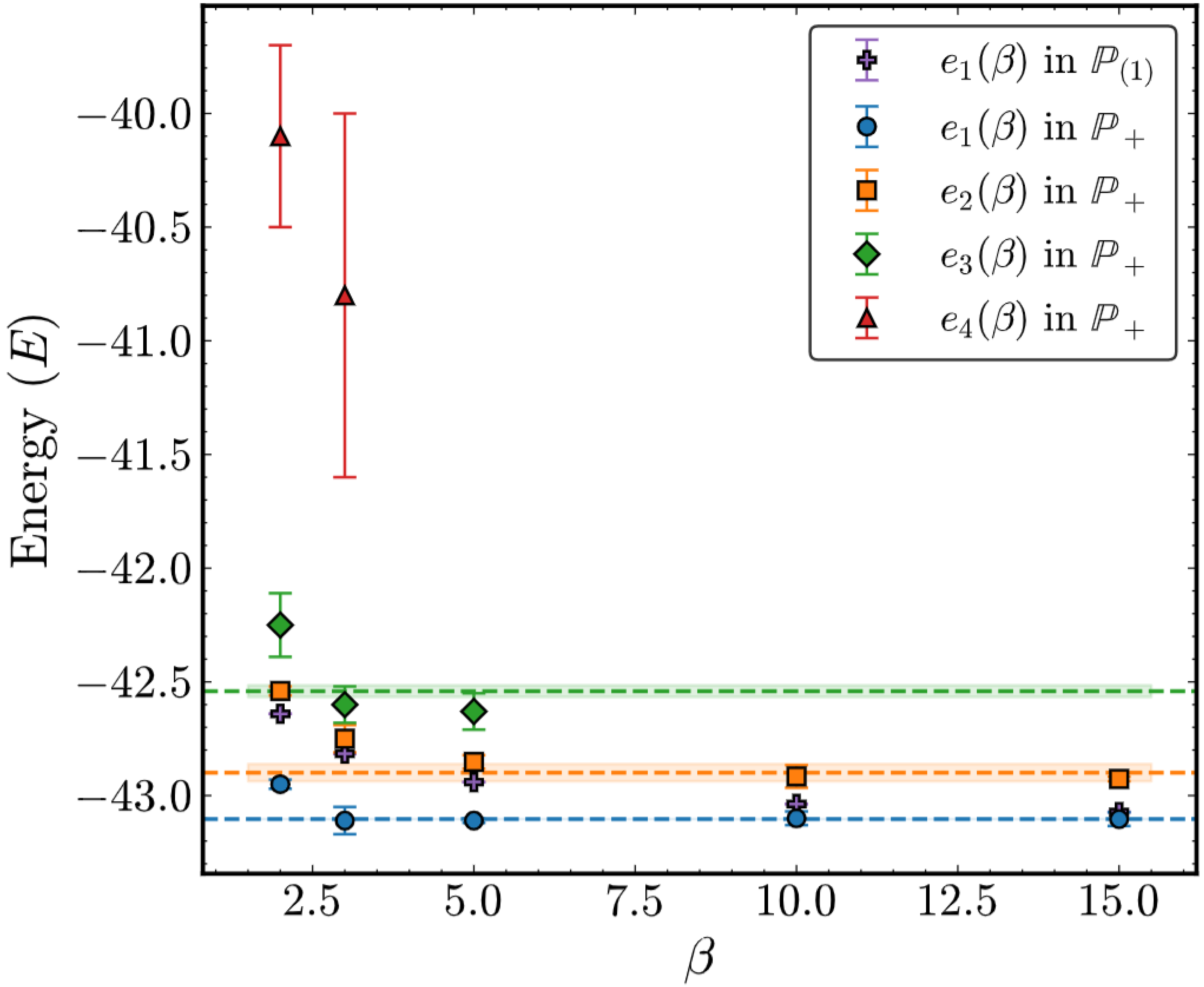
The four low-lying excited states are accessed at very low beta!
(no sign problem here.)

The 8x8 Unfrustrated Heisenberg Limit



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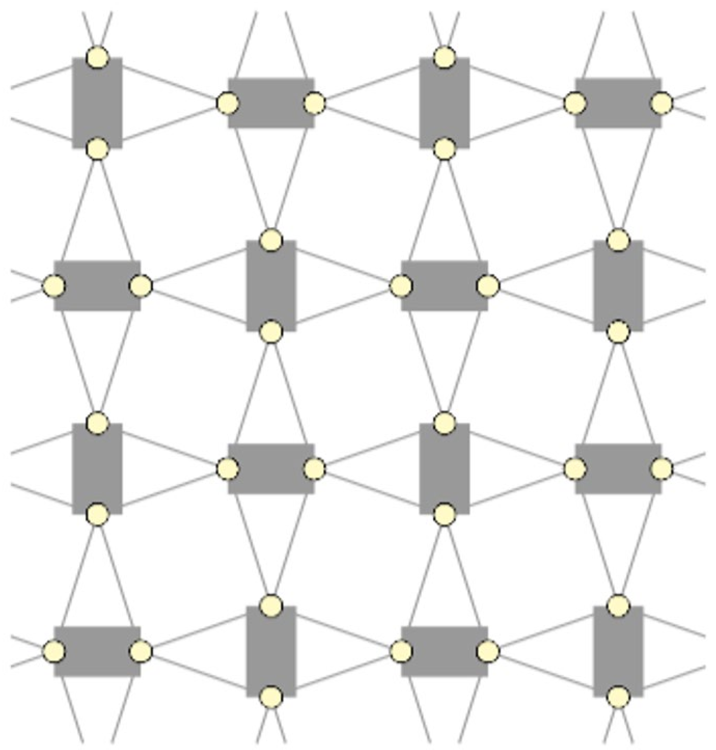
- PDMS converges, within errors, to low energy sector at rather small betas.
- Judicious increase in projection subspace dimension helps for rapid convergence.

Projection Subspaces Captures the Physics : II

Dimer



$$\hat{Q} = \prod_i (2s_i^x)$$

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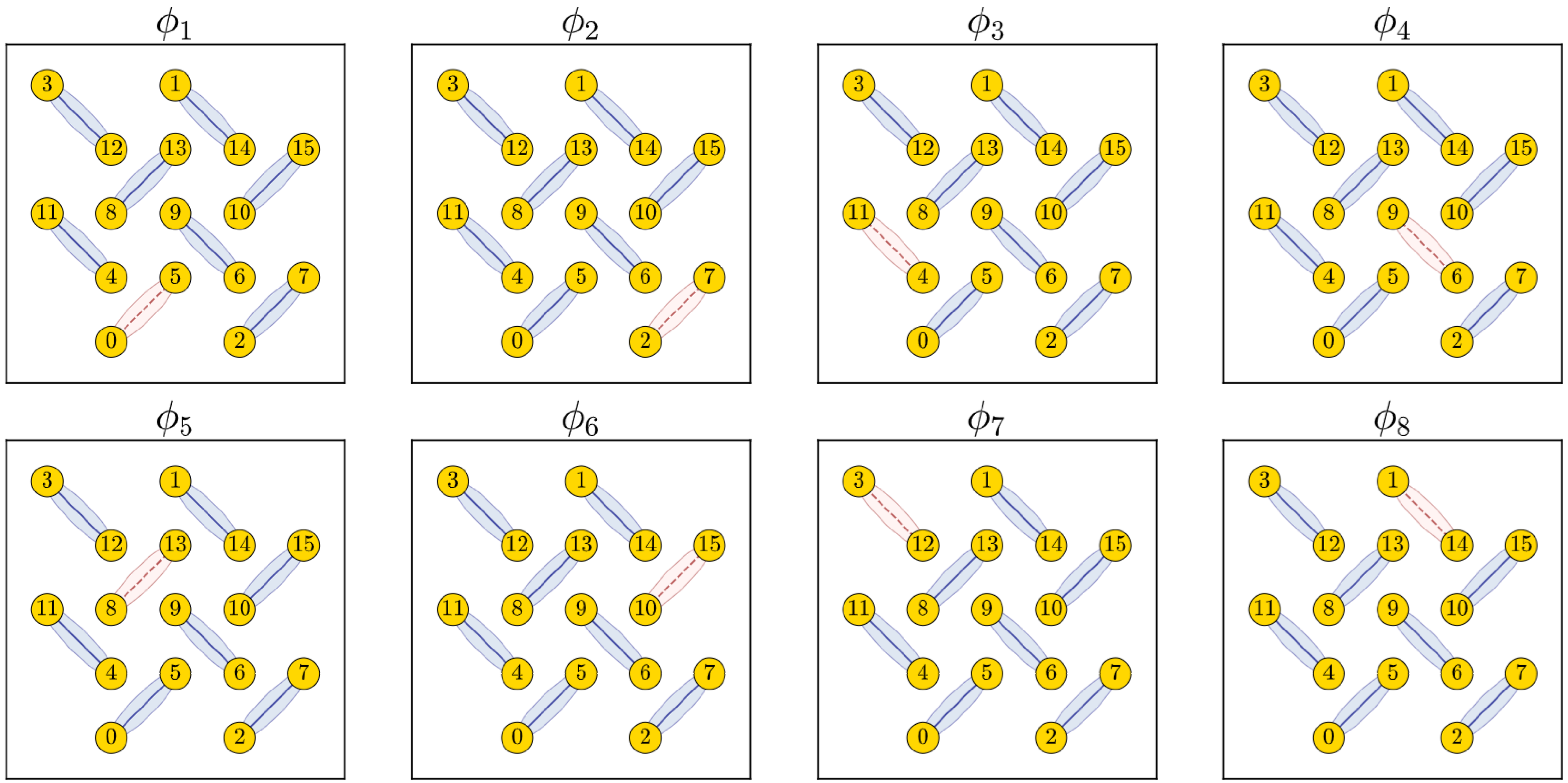
Two projection sectors with +1 and -1 eigenvalues!

P- (phi states)



These states are motivated by the physics of the dimerized phase!

The Frustrated SSM in the Dimer Phase: J'/J=0.5

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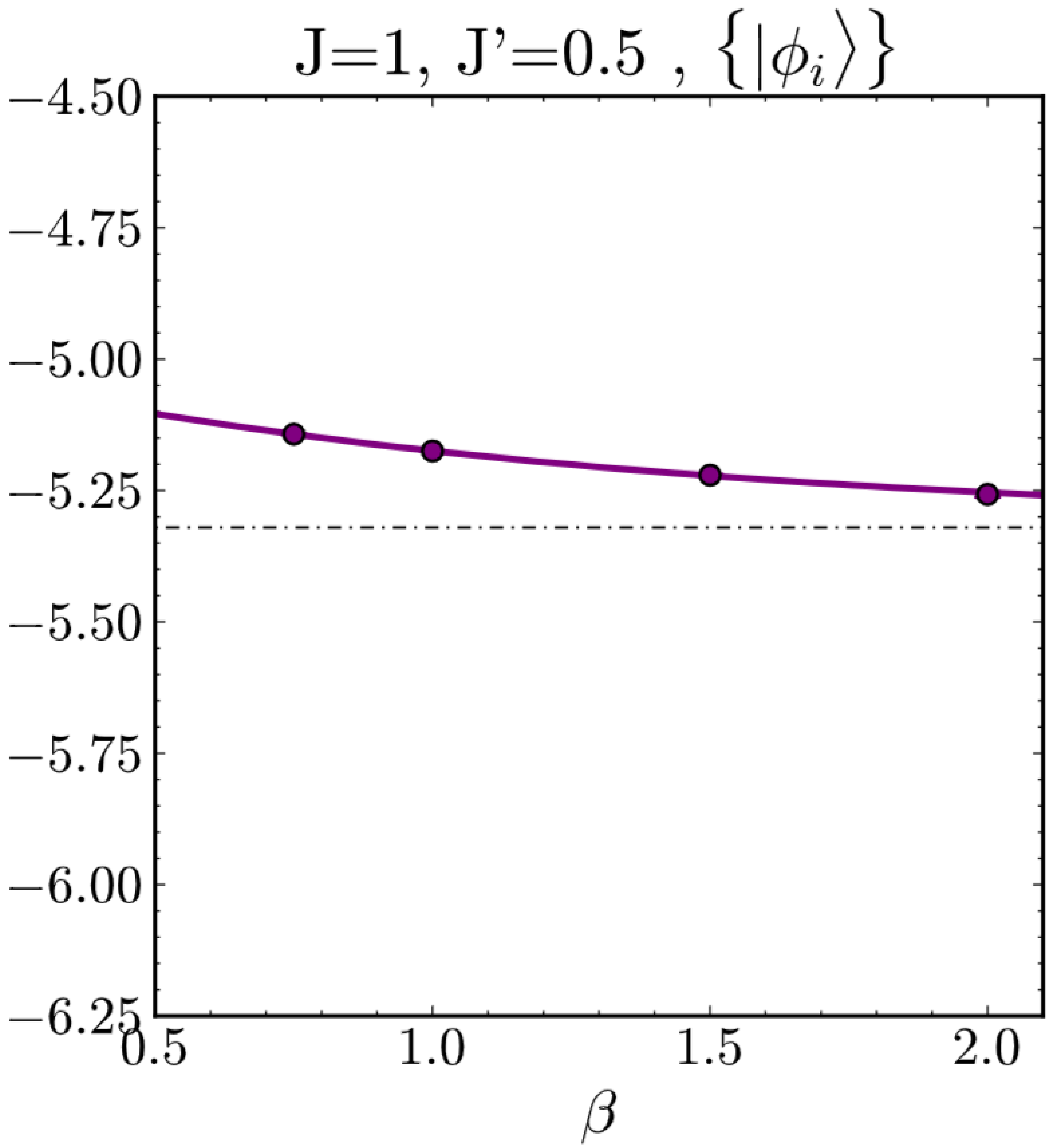

- PDMS converges close to the expected projected lowest energy state at beta=2!
- Notice that the 8 phi states are effectively degenerate…part of the physics of SSM.

The SSM Lattice in the Highly Frustrated Phase: J'/J=0.75

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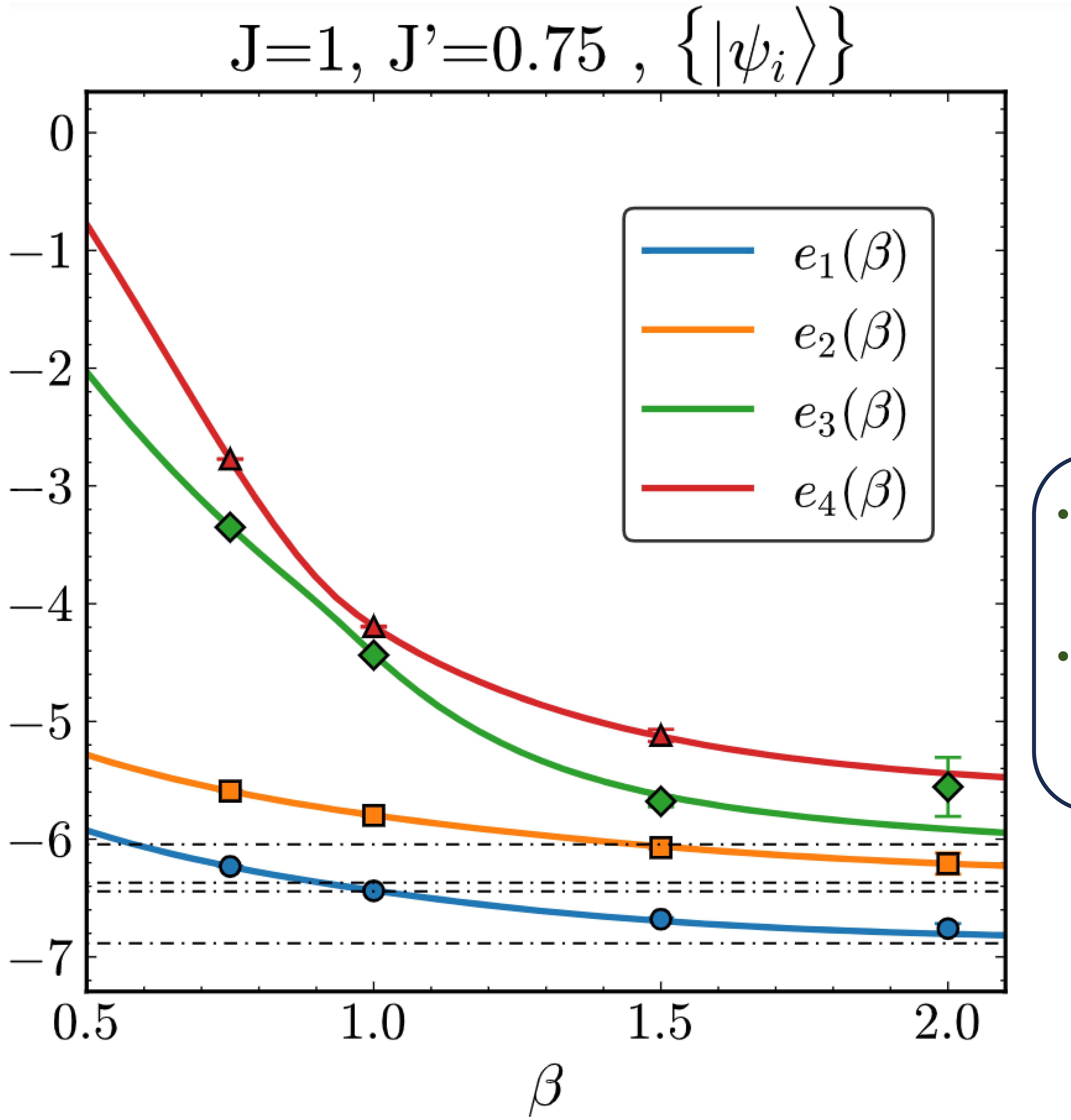

- Notice that the PDMS results agree perfectly with the exact solid lines $E(\beta)$!
- Question: Can we find better projection basis to describe this part of the phase diagram?

At J'/J = 0.75, the anti-ferromagnet-like P+ (psi states) subspace turns out to do great to access the ground state! Other higher states remain unresolved…

The 8x8 Unfrustrated Heisenberg Limit: Energy Gap

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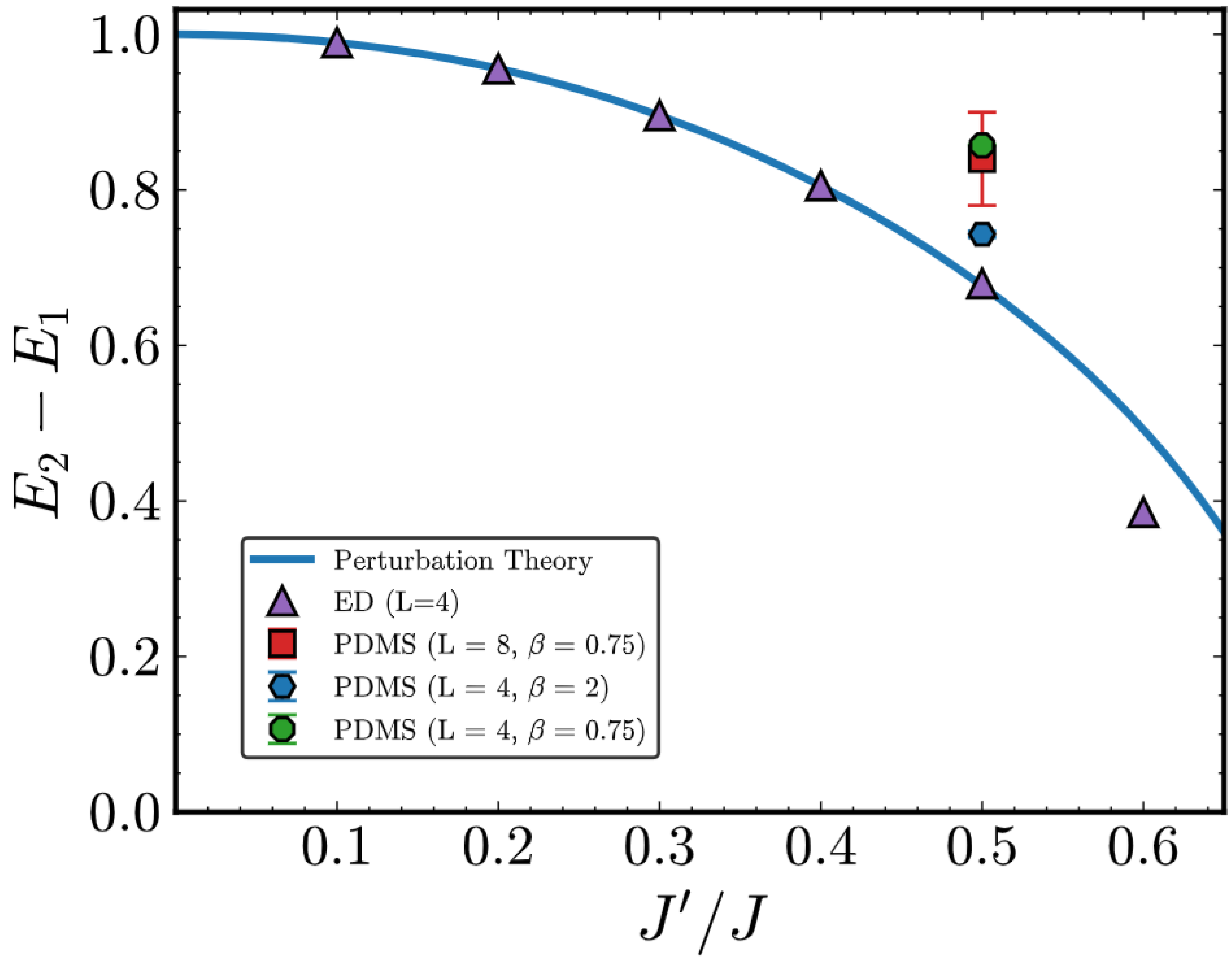
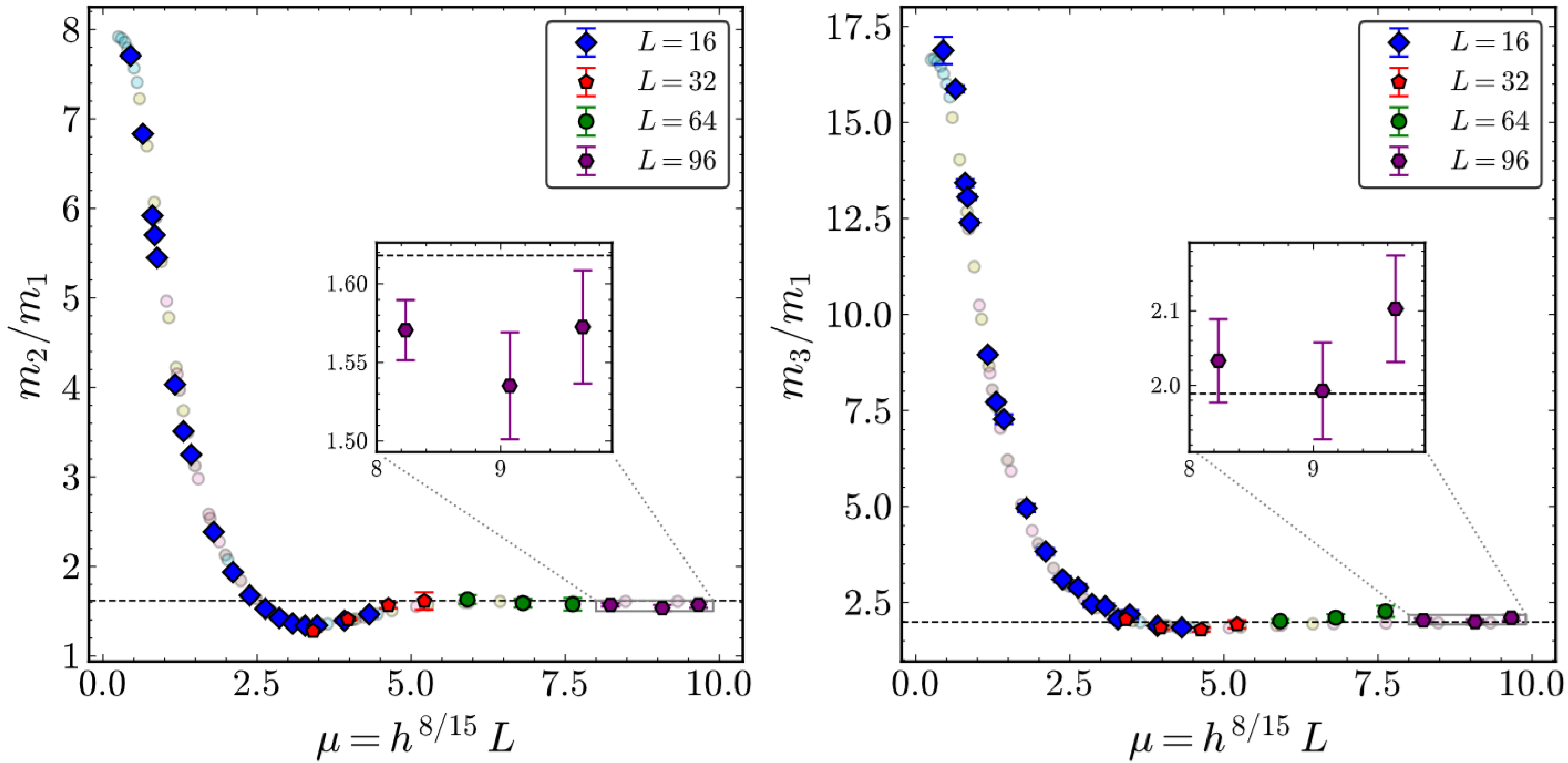


- **Non-perturbative effects remain small up to J'/J~0.5,**

Bonus Example: 1D-TFIM with Longitudinal Field at Criticality



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Details in here

- **Great accuracy in resolving the first two mass ratios of the massive E8 QFT spectrum!**

(previously QMC have not obtained to the best of our knowledge; however, tensor network studies have obtained these before.)

Conclusions and Future Works

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- PDMS is a new approach **to mitigate the sign problem and access low-energy excited states**.

- The method is crucially dependent on the choice of projection subspace!
  - **Can symmetries help us find a nice projection subspace?**
  - **Can perturbation theory guide us?**
  - **Can experiments or physical insight guide us? PVB, RVB states?**

- How far can we push the sign problem? Can we find **a general approach to converge at low beta for some class of problems** before sign problem becomes prohibitive?

- What if we use a different computational basis? **Dimer basis vs Sz basis**?

- Broadly, **can we capture the physics of a system by looking at specific subsystems**?

Acknowledgements

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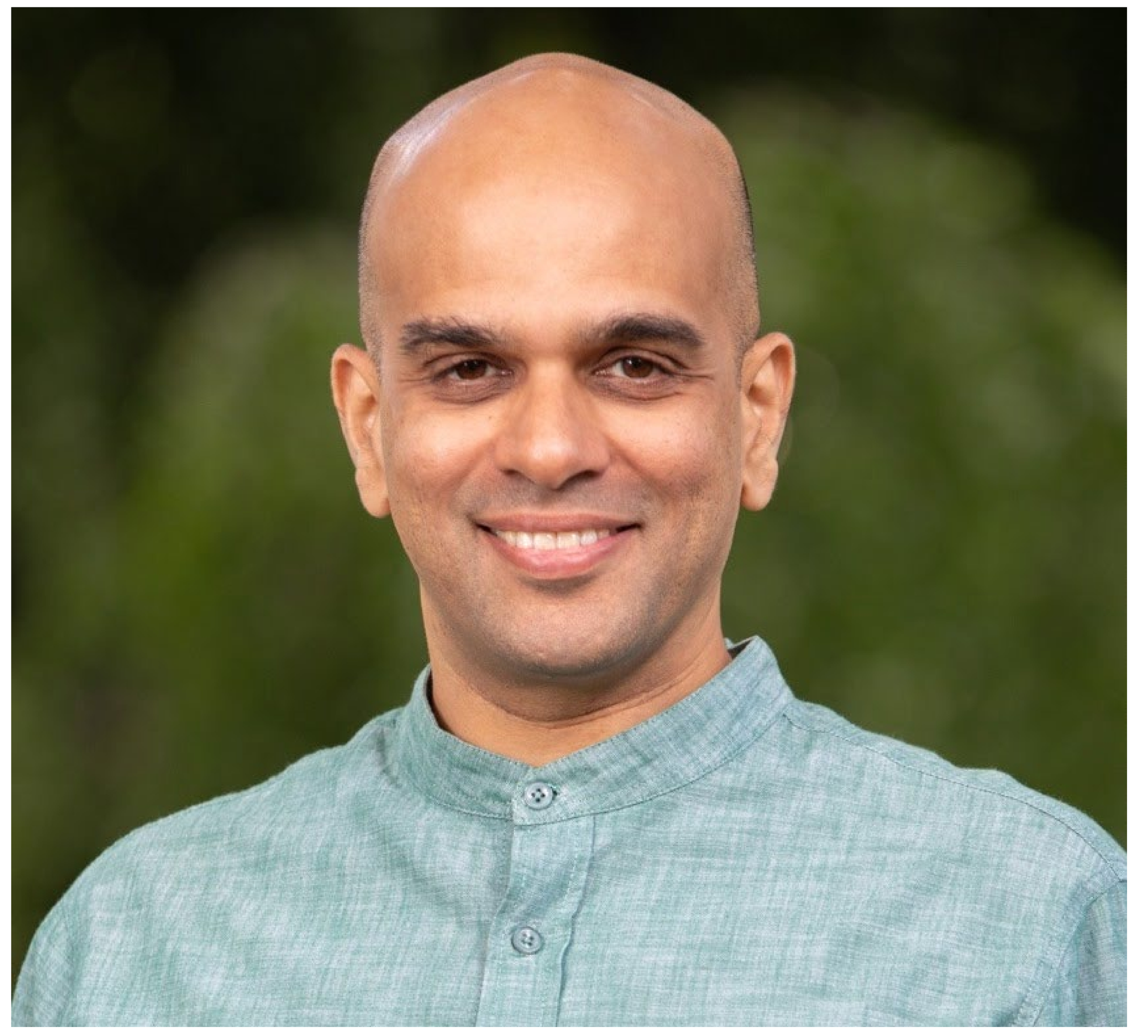

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Hansen Wu

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Thank you for your attention.

Questions?